



Chemical Ecology

Making “scents” of how plant volatiles influence agriculturally important insects: a review

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Plants emit hundreds, if not thousands, of different volatile chemical compounds, although the function of most individual volatiles remains elusive. Individual volatiles, as well as blends of many chemicals, are likely multifunctional in regulating plant interactions with different groups of insects, including herbivores, natural enemies, and pollinators. However, research on these insect groups has historically been siloed, limiting our understanding of connections between different volatile-mediated ecological processes and how to apply this knowledge to agroecosystems. Here, we review recent literature on volatile multifunctionality in mediating plant interactions with insect herbivores, natural enemies, and pollinators. Ultimately, we propose that future research shifts towards a holistic approach in the study of volatile-mediated interactions between plants and insect communities. By elucidating how specific volatiles, chemical classes, and blends regulate behaviors across different groups of insects, we will uncover new semiochemical tools for controlling pests and protecting beneficial insects in agroecosystems.

Keywords: herbivore-induced plant volatiles, volatile organic compounds, pest management

Introduction

A longstanding goal in chemical ecology is to identify semiochemicals that improve the management of agriculturally important insects (Pickett et al. 1997, Smart et al. 2014). Plant volatiles are key semiochemicals known to regulate ecological interactions between crop plants and insects (Szendrei and Rodriguez-Saona 2010, Hare 2011, Dicke and Lucas-Barbosa 2020). Agriculturally important insects range from herbivorous pests—which feed on crop plants and cause billions of dollars in crop damage each year—to beneficial insects. Beneficial insects include natural enemies (ie predators, parasitoids) that consume, parasitize, or infect insect pests (Lacey et al. 2015, Hajek and Eilenberg 2018, Garcia et al. 2020) and pollinators, which facilitate reproduction for 85% of flowering plants including 87 globally important crops (Klein et al. 2007, Ollerton et al. 2011). Herbivores, natural enemies, and pollinators are all known to detect and respond to plant volatiles (Takabayashi and Shiojiri 2019). To modulate the behaviors of different insect groups, ecological theory suggests individual plants emit diverse blends of volatile compounds (Kessler and Kalske 2018, Wetzal and Whitehead 2020, Whitehead et al.

2021). Volatile diversity is thereby predicted to positively correlate with the diversity of interactions between an individual plant and different insects. However, most empirical research has focused on how plant volatiles influence interactions with one or two insects, neglecting the complexity and diversity of insect communities interacting with crop plants. With minimal knowledge of how volatiles regulate insect community-wide interactions with plants, applications in agroecosystems remain limited.

Volatile organic compounds, which plants constantly emit into the atmosphere, are carbon-containing chemicals with high vapor pressures and often categorized by their biosynthetic pathway of origin, including terpenoids, phenylpropanoids/benzenoids, fatty acid derivatives, and amino acid derivatives (Baldwin 2010, Dudareva et al. 2013). Plants emit blends of volatiles containing a diverse array of compounds, and emerging evidence has revealed the temporal patterns and specificity of blends across different plant tissues, genotypes, and species (Clavijo McCormick et al. 2012, McCormick et al. 2014, Russavage et al. 2024, Thompson et al. 2025a). For instance, flowers produce volatile blends that often overlap but

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sometimes diverge in chemistry with the blends of vegetative volatiles emitted by leaves and roots (Knudsen et al. 2006). Adding further complexity, plants emit volatile blends under normal physiological conditions (“constitutive volatiles”) but also modify volatile emissions in response to environmental stressors (“induced volatiles”). Induced plant volatiles can provide protection against different abiotic and biotic stressors either directly or through indirect effects (eg activation of molecular signaling pathways), improving plant stress resistance and resilience (Holopainen and Gershenzon 2010, Possell and Loreto 2013, Zuo et al. 2025). Whether constitutive or induced, distinctive plant volatile blends encode detailed ecological information, which foraging insects can use to assess food resources and habitats.

Foraging insects interpret plant volatile emissions to distinguish host identity, habitat quality, and the presence of other insects (Bruce and Pickett 2011, Aartsma et al. 2017, Grunseich et al. 2019). In general, insects are either attracted to and navigate towards preferred plant volatiles, or repelled by and move away from deterrent plant volatiles. To manipulate the behaviors of agriculturally important insects, harnessing plant volatiles may provide an environmentally friendly management strategy for improved ecosystem services and crop production. In recent decades, calls for sustainable insect management strategies in agroecosystems have intensified (Brzozowski and Mazourek 2018, Hawes et al. 2021). Overuse of agrochemicals to control pest populations has led to accumulating incidences of insecticide resistance and environmental health concerns (Devi et al. 2022, Pu and Chung 2024). Simultaneously, managing beneficial natural enemies and pollinators has become increasingly challenging given the growing prevalence of dynamic agroecological stressors (ie agrochemicals, variable weather patterns, invasive species; Thomson et al. 2010, Welch and Harwood 2014, Goulson et al. 2015, Potts et al. 2016). Therefore, developing a deeper understanding of how plant volatiles regulate insect behaviors could equip growers with additional tools to manage insect pests and safeguard beneficial insects in a changing landscape.

Given the importance of insect herbivores, natural enemies, and pollinators to agricultural production and the human food supply, we review how plant volatiles mediate behaviors of multiple groups of insects simultaneously. First, we examine how volatiles shape plant interactions with herbivores and natural enemies, then we consider herbivores and pollinators, and finally natural enemies and pollinators. We focus on what we define as “positive” and “negative” outcomes of volatile-mediated interactions, corresponding to whether plant volatiles modify insect behaviors to enhance or harm agricultural production. Ultimately, we emphasize the need to leverage “positive” plant volatiles that promote natural enemies and pollinators while repelling insect pests. We conclude with a discussion of emerging applications in agricultural systems and perspectives on future research directions to propel the use of plant volatiles for agricultural insect management.

Volatiles Mediate Plant Interactions With Herbivores and Natural Enemies

Insect feeding damage alters plant volatile emissions qualitatively and quantitatively, resulting in a unique blend of herbivore-induced plant volatiles (HIPVs). Emitting HIPVs can

“directly” defend many different crop plant species from insect pests by repelling, deterring, or intoxicating herbivores (De Moraes et al. 2001, Veyrat et al. 2016, Ray et al. 2020). A major breakthrough over the past four decades was the discovery that natural enemies also respond to HIPVs and capitalize on such signals to locate prey or hosts (Dicke and Sabelis 1987, Turlings et al. 1990, Heil 2008). Plants can defend themselves “indirectly” by attracting natural enemies to kill herbivores through the emission of HIPVs (for more details, see Mumm and Dicke 2010, Aljibory and Chen 2018, Niu et al. 2024). This attraction is well documented in many cropping systems, including apple (El-Sayed et al. 2018), cabbage (Poelman et al. 2009), cranberry (De Lange et al. 2019), cucumber (Kappers et al. 2010), lima bean (Ballhorn et al. 2008), maize (Grof-Tisza et al. 2024, Wang et al. 2025), rice (Lou et al. 2006), sorghum (Russavage et al. 2024), and tomato (Silva et al. 2018). However, it is important to note that while HIPVs are known to attract natural enemies, relatively few studies have demonstrated that increased natural enemy attraction translates into enhanced plant fitness. As a result, many questions remain regarding the definitive functions of HIPVs in plant defense. Nevertheless, based on the predicted roles of HIPVs in direct and indirect plant defense, we highlight examples of plant volatiles that simultaneously influence behaviors of herbivorous insect pests and natural enemies. We define “positive” outcomes as advantageous to agricultural production, including examples of plant volatiles repelling herbivorous insect pests while increasing natural enemy activity, assuming such interactions improve plant growth and crop yield—although few studies have measured such effects on yield. In contrast, we define “negative” outcomes as examples of plant volatiles attracting insect pests while repelling natural enemies (Fig. 1).

Positive Outcomes—Repel Herbivores, Attract Natural Enemies

Plant volatiles can have dual ecological functions, including repelling insect herbivores while also attracting natural enemies. Occasionally, a single volatile compound can underlie such duality. For example, (Z)-3-hexenol emission alone attracts parasitoids (Wei et al. 2007) and also plays an important role in repelling insect pests (Wei and Kang 2011). However, in many other cases, blends of multiple plant volatile compounds, often emitted in specific ratios to one another, are needed to elicit behavioral responses in insect herbivores and natural enemies (Clavijo McCormick et al. 2012, Zhou and Jander 2022). A wide taxonomic range of plant species rely on plant volatile duality for direct and indirect defense, including grasses (ie maize [*Zea mays* L.], Wang et al. 2025), herbaceous plants (ie tomato [*Solanum lycopersicum* L.], Yang et al. 2020, cucumber [*Cucumis sativus* L.], Grunseich et al. 2020, and evergreen trees (ie tea [*Camellia sinensis* (L.) Kuntze], Jiao et al. 2022, *Aquilaria sinensis* (Lour.) Gilg) Qiao et al. 2018). Plants likely benefit most from volatile duality in direct and indirect defense when the threat of herbivory is severe and natural enemies substantially suppress herbivores (Fatouros et al. 2014). However, we still lack a comprehensive understanding of which crop plant species rely on volatile duality for direct and indirect defense, as well as how many different types of pests are thwarted by such defenses. Additionally, artificial selection through crop domestication or extensive breeding can modify volatile emissions both quantitatively and qualitatively

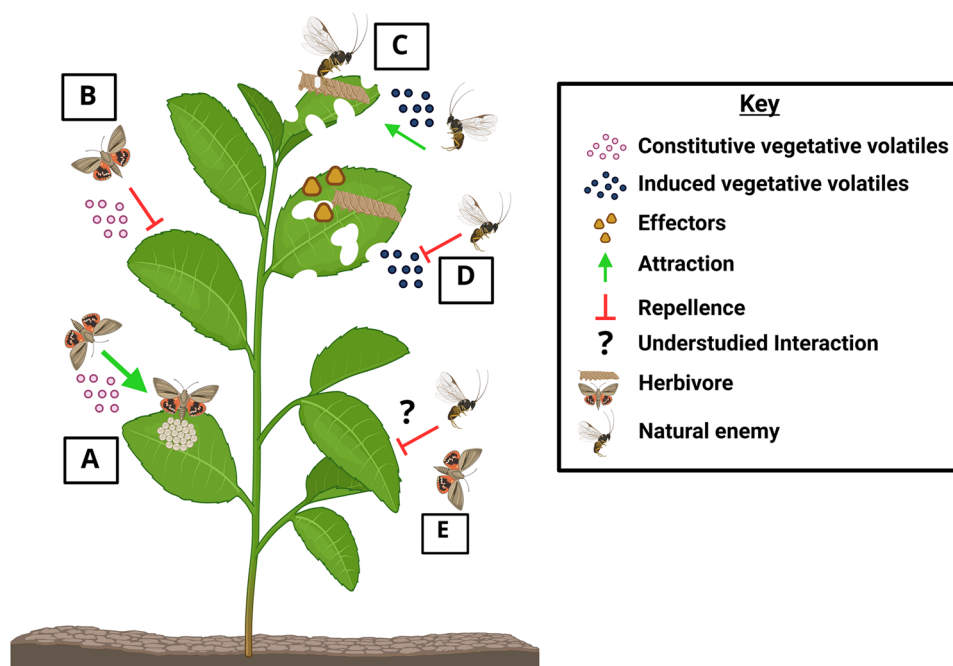


Fig. 1. Volatile-mediated plant interactions with insect herbivores and natural enemies depicting: (A) herbivore host plant location, (B) direct defense, (C) indirect defense, (D) herbivore defense manipulation, and (E) repellency of both herbivores and natural enemies. The key identifies constitutive (light circles) or induced (dark circles) vegetative volatiles, herbivore salivary effectors (triangles), insect attraction (pointed arrows) or repellence (blunted arrows), whether the interaction is understudied in the literature (question mark), and insect functional group (herbivores or natural enemies).

(Köllner et al. 2008, Rowen and Kaplan 2016), highlighting the need for greater consideration of volatile duality in crop breeding for direct and indirect defense.

Certain plant volatiles attract natural enemies as well as insect herbivores. In such cases, plant recruitment of natural enemies for indirect defense may result in the unavoidable consequence of attracting more herbivores. For instance, apple (*Malus domestica* (Suckow) Borkh.) trees infested with pandemis leafroller moths (*Pandemis pyrusana* Kearfott) attract green lacewing predators, as well as more adult moths (El-Sayed et al. 2018). Relatedly, following feeding damage from western corn rootworm (*Diabrotica virgifera virgifera* LeConte), maize (*Z. mays*) roots emit increased levels of (*E*)- β -caryophyllene into the rhizosphere, enhancing entomopathogenic nematode attraction (Rasmann et al. 2005), but (*E*)- β -caryophyllene also attracts more rootworm pests (Robert et al. 2012). Such scenarios of dual recruitment through volatile emissions may arise when the benefit of attracting natural enemies outweighs the drawbacks of heightened herbivory. Alternatively, plants may also rely on other defensive strategies in conjunction with HIPV-mediated indirect defense. For instance, in addition to indirect defense, maize plants also activate vigorous root regrowth to overcome pest feeding and can even overcompensate for herbivory (Robert et al. 2015). Thereby, dual recruitment of herbivores and natural enemies may not ultimately have a negative impact on maize plants. It is also important to note that HIPVs may attract other pests or antagonists beyond the initially attacking herbivore species. For instance, citrus roots emit HIPVs after weevil damage, which are attractive to entomopathogenic nematodes, but these HIPVs also attract plant parasitic nematodes (Ali et al. 2011). Indirect defense in citrus may protect against attacking herbivores, but HIPVs appear to leave roots vulnerable to other antagonists. Determining whether HIPVs recruit natural enemies alone or in

conjunction with specific pest species is critical for determining how to best incorporate plant volatiles into pest management strategies.

Negative Outcomes—Attract Herbivores, Repel Natural Enemies

In certain cases, plant volatiles attract herbivores but repel natural enemies (D'Alessandro and Turlings 2006, Snoeren et al. 2010, Ye et al. 2018). For instance, herbivorous spider mites (*Tetranychus urticae* C.L. Koch) are attracted to foliar HIPVs emitted by cucumbers (*C. sativus*), while predatory mites (*Phytoseiulus persimilis* Evans) avoid four specific cucumber HIPVs: (Z)-3-hexenyl acetate, farnesol, methyl salicylate, and (*E*)- β -ocimene (Pallini et al. 1997, Kappers et al. 2010). Emitting volatiles that attract herbivores and repel natural enemies appears problematic for plants, which, as a result, may sustain higher levels of insect feeding damage. One intriguing possibility for why this may occur is that herbivorous pests manipulate HIPVs for their own benefit to recruit more conspecifics or reduce detectability from natural enemies (Rodriguez-Saona et al. 2003). For example, tomato fruitworm caterpillars (*Helicoverpa zea* Boddie) secrete salivary enzymes that manipulate tomato (*S. lycopersicum*) leaves to close stomata and thereby suppress HIPV emissions that typically recruit natural enemies (Lin et al. 2021, 2022). Similarly, microbial endosymbionts in the saliva of pea aphids (*Acyrtosiphon pisum* Harris) reduce total volatile emissions of fava beans (*Vicia faba* L.), which decreases the attraction of parasitoid wasps (Frago et al. 2017). In addition to herbivore manipulation, another possible explanation for natural enemy avoidance of plant volatiles is that natural enemies may rely on HIPVs to gauge competition and hyperparasitism (Poelman et al. 2013). For instance, Brussels sprout (*Brassica oleracea* L.) plants attacked by non-parasitized caterpillars emit higher quantities of terpenes, such as

limonene, α -thujene, and 1,8-cineole, which the parasitoid *Cotesia glomerata* (L.) prefers to volatiles of plants treated with parasitized caterpillars (Fatouros et al. 2005). Relatedly, other studies have confirmed that when compounds are missing from HIPV blends, there is a decrease in natural enemy activity (Rasmann et al. 2005, D'Alessandro et al. 2009). Emission of HIPVs can negatively impact host or prey finding for natural enemies while simultaneously boosting herbivorous pest attraction, but why such volatile-mediated interactions may evolve in plant populations remains underexplored. Ultimately, there are only a handful of studies to date that have demonstrated natural enemy avoidance of HIPVs and more work is needed to determine if these volatile-mediated interactions are common or rare.

In other cases, plant volatiles simultaneously repel insect herbivores and natural enemies. Plants may avoid recruiting insects altogether if the risk of herbivory is very high, indicating an extreme bet-hedging strategy of plant defense. For instance, isoprene repels foraging Lepidopteran herbivores (Laotawornkitkul et al. 2008) and natural enemies (Loivamäki et al. 2008). While certain volatiles like isoprene may serve as a potent defense, the drawbacks for plants of repelling herbivores as well as natural enemies are important to consider, particularly when deploying volatiles for insect management in agroecosystems. It is also worth noting that some insects are both predatory and herbivorous (ie omnivores; Blubaugh 2023). If omnivores are abundant, plants may benefit most by not attracting any insects to minimize the possible threat of consumption. In addition to a high risk of herbivory underlying volatile-mediated interactions, exposure to plant volatiles can also modify insect herbivore physiology and immunity, which can, in turn, affect natural enemies (Hu et al. 2021). For example, in the maize (*Z. mays*) system, leaf emissions of indole not only repel insect herbivores but also modify the odor profile of herbivores themselves (Ye et al. 2018). While natural enemies are not repelled by indole alone, the combination of indole and altered herbivore odors repels natural enemies (Ye et al. 2018). Relatedly, in belowground systems, exposing herbivores to common root HIPVs, including pregeijerene, linalool, β -ocimene and α -pinene, modifies herbivore immunity and reduces the survival of endoparasitic entomopathogenic nematodes (Gaffke et al. 2022). Whether plant volatiles directly repel natural enemies or indirectly modify insect herbivore physiology to negatively impact natural enemies, the overall effect on biological control is potentially critical and merits future study.

Volatiles Mediate Plant Interactions With Herbivores and Pollinators

To facilitate sexual reproduction, flowers emit volatile blends that are attractive to insect pollinators. However, insect herbivores also respond to floral volatiles, particularly herbivores that feed on flowers (ie florivores) or certain pollinators that also function as herbivores across different life stages (ie Lepidopteran adults and larvae; Kessler et al. 2010, Page et al. 2014). While floral volatiles are emitted constitutively under normal physiological conditions, insect herbivory can also alter floral volatiles (Lucas-Barbosa et al. 2011, Rusman et al. 2019). Herbivore-induced changes to floral volatiles may alter their attractiveness to different insects. Therefore, plants must

balance competing demands when activating defenses and emitting floral volatiles: luring pollinators while also avoiding herbivores. Over evolutionary time, the strength of plant interactions with pollinators and herbivores can influence floral volatiles and ultimately contribute to plant fitness (Ramos and Schiestl 2019), but we have little understanding of such processes in agricultural crops or how to apply this understanding to crop breeding efforts and agricultural insect management. Here, we summarize examples of floral and vegetative volatiles mediating interactions between insect pollinators and herbivores. We define “positive” outcomes as instances of plant volatiles repelling herbivorous insect pests while attracting beneficial pollinators or improving pollination services (ie increasing fruit or seed set). We define “negative” outcomes as examples of plant volatiles increasing pest attraction while reducing the benefits provided by pollinators (Fig. 2).

Positive Outcomes—Repel Herbivores, Attract Pollinators

Plants strategically balance interactions with herbivores and pollinators through dynamic volatile emissions, simultaneously maximizing reproductive output while reducing excessive herbivory. For example, plants may facilitate such dual interactions through temporal variation in the release of specific volatiles, which may be most effective when herbivores and pollinators are active at different times (ie crepuscular herbivores and nocturnal pollinators; Kessler et al. 2019). In addition to temporal variation, plants may also maximize the probability of successful sexual reproduction by emitting distinct volatiles from different tissues, such as leaves and flowers or petals and sepals (Euler and Baldwin 1996). In an extreme act to avoid herbivory while maintaining sufficient pollination, wild tobacco, *Nicotiana attenuata* (Torr. ex S. Watson), can modify its flowering phenology. In response to heavy bouts of hawkmoth (*Manduca* spp.) caterpillar herbivory, *N. attenuata* shifts from producing nocturnal, hawkmoth-pollinated flowers, to diurnal flowers, which emit different levels of benzyl acetone and attract day-active hummingbirds (Kessler et al. 2010). Together, these examples demonstrate the complex emission patterns of plant volatiles—and in extreme cases, flexible flowering phenology—that serve as critical tools for plants to maintain a delicate equilibrium between minimizing the costs of herbivory and maximizing the benefits of pollination. An important caveat here is that, to date, most examples of plants repelling herbivores while attracting pollinators through volatile emissions come from a single study system: wild tobacco, *N. attenuata*. However, it is highly likely that other plant species utilize volatiles in similar ways, indicating that future studies on plant–herbivore–pollinator interactions should prioritize additional plant families, as well as agricultural crop species.

In other instances, certain volatiles simultaneously attract both pollinators and herbivores. For example, butternut squash (*Cucurbita*) blossoms produce 1,2,4-trimethoxybenzene, recruiting specialist pollinators and herbivores: squash bees (*Eucera pruinosa* Say) and striped cucumber beetles (*Acalymma vittatum* Fabricius), respectively (Andrews et al. 2007). Striped cucumber beetles (*A. vittatum*) are a major pest in squash agroecosystems, indicating potentially detrimental effects for squash plants that attract both pollinators and herbivores. However, not all herbivores are harmful to plant reproduction. In some cases, plants can rely on a combination of

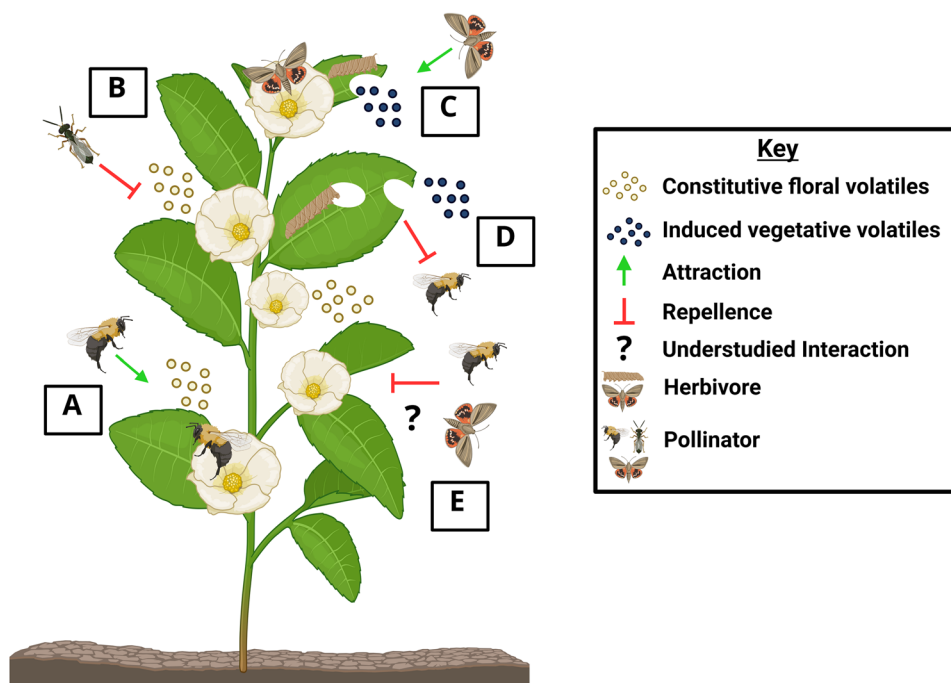


Fig. 2. Volatile-mediated plant interactions with insect herbivores and pollinators depicting: (A) pollination, (B) maximized pollination by repelling visitors from already-pollinated flowers, (C) attraction of herbivores that also pollinate flowers, (D) herbivory-mediated repellence of pollinators, and (E) repellence of both herbivores and pollinators. The key identifies: constitutive floral (light circles) or induced vegetative (dark circles) volatiles, insect attraction (pointed arrows) or repellence (blunted arrows), whether the interaction is understudied in the literature (question mark), and insect functional group (herbivores or pollinators).

pollinating and herbivorous insects for successful reproduction. For example, the orchid *Dichaea pendula* ((Aubl.) Cogn.) attracts both pollinators and herbivores by emitting 2-methoxy-4-vinylphenol, which facilitates sexual reproduction with orchid bees (Euglossini) and also attracts herbivorous weevils. Orchid bees are not very abundant in the landscape and weevil florivory initiates self-pollination in *D. pendula* (Nunes et al. 2016), indicating that these orchids rely on herbivorous weevils for reproduction when pollinators are rare. In addition to herbivores playing a direct role in plant reproduction, herbivores can also indirectly influence reproduction through plant emission of HIPVs. Pollinators leverage the chemical information from HIPVs for their own benefit to locate plants with high-quality flowers or abundant floral resources. For instance, white campion plants (*Silene latifolia* Poir.) emit higher amounts of (Z)-3-hexenyl acetate and β -ocimene when infested with herbivores (*Spodoptera littoralis* Boisduval) and these plants also produce more fruits than open-pollinated control plants, suggesting that increased emission of HIPVs enhances nocturnal pollinator attraction (Cozzolino et al. 2015). When pollen-limited, bumble bees (*Bombus terrestris* L.) may even take on the role of damage-inflicter, biting holes into *S. lycopersicum* and *Brassica nigra* (L. Al-Shehbaz) leaves to induce flowering, increasing available floral resources (Pashalidou et al. 2020). Overall, floral volatiles can be multifunctional, attracting insect pollinators and herbivores, but it is not well understood how widespread these volatile-mediated interactions are across plant families. Furthermore, crop plants may experience greater herbivore pressure than wild plants, making it more detrimental for agricultural crop plants to draw in both herbivores and pollinators simultaneously. However, crop plants are less

resource-limited and thereby may tolerate more herbivore damage before experiencing yield loss, which is important to consider in future research on crop plants and their wild relatives.

Negative Outcomes—Attract Herbivores, Repel Pollinators

When herbivory obstructs pollination, plants are unable to reproduce and in turn, are unable to produce fruits and seeds, many of which are necessary for the human food supply. Herbivory may induce the emission of new or different volatiles (Vega-Polanco et al. 2020), which can have negative consequences for plant-pollinator interactions (Kessler and Halitschke 2009, Kessler et al. 2011). Conversely, herbivory can reduce the number of volatiles a plant produces or reduce the quantity of specific volatiles that attract pollinators (Pareja et al. 2012, Schiestl et al. 2014). Specifically, herbivore-induced changes to floral volatiles can lead to fewer or shorter pollinator visits (*Solanum peruvianum* L., Kessler and Halitschke 2009, Kessler et al. 2011, *S. rostratum* Dunal, Vega-Polanco et al. 2020, *Brassica rapa* L., Schiestl et al. 2014, Moreira et al. 2024, *B. nigra*, Bruinsma et al. 2014, Lucas-Barbosa et al. 2016) reduced seed set (*S. peruvianum*, Kessler et al. 2011, *Cucurbita pepo* (L.) var. *texana*, Theis and Adler 2012), and lower seed quality (*B. rapa*, Moreira et al. 2024). Changes in floral volatiles may depend on herbivore feeding guild or dietary specialization. For example, phloem-feeding aphids in general, and specialist aphids, specifically, elicit stronger plant and pollinator responses than chewing caterpillars (*B. rapa*, Moreira et al. 2024, *Sinapis alba* L., Pareja et al. 2012). However, Vega-Polanco et al. (2020) reported that florivore-induced changes to floral volatiles in *Solanum rostratum* (Dunal) impacted the frequency and duration of pollinator visits, but

had no corresponding change in fruit or seed production. This could suggest that the pollination requirements of this plant are relatively low (ie quick, infrequent visits are sufficient for pollination) or that secondary selfing mechanisms make up for reduced pollinator activity. It is also possible that herbivore-induced changes to floral volatile blends could mask the location of flowers to pollinators, although evidence of these processes is currently limited.

To our knowledge, there are no examples of volatiles simultaneously repelling foraging herbivores and pollinators in the published literature. Despite this knowledge gap, it is clear that the relationship between plant volatiles, herbivores, and pollinators is nuanced. While herbivory often modifies floral volatiles and decreases pollinator attraction (see above), in some cases, herbivory modifies floral volatiles without any negative consequences on pollination. For example, herbivory on *B. nigra* flowers and leaves altered floral volatiles, but did not change floral attractiveness to *Pieris brassicae* (L.) butterflies (Bruinsma et al. 2014, Lucas-Barbosa et al. 2016). For other plants, herbivory does not alter floral volatiles (Reisenman et al. 2013) or pollination outcomes (Zangerl and Berenbaum 2009). However, additional research is needed to comprehensively understand when herbivory changes floral volatiles and the factors shaping positive, negative, or neutral interactions with pollinators. Interestingly, plant density may further influence plant–herbivore–pollinator interactions. In tall goldenrod, *Solidago altissima* (L.), ramet density influenced the effects of herbivory on floral volatiles: when growing in low density, herbivory did not affect floral volatile emissions, but when grown in high density, herbivory suppressed floral volatiles; however, seed set did not vary by density (Herschberger et al. 2024). Since agricultural crops are often grown in high density, it is important for future work to consider how crop density

affects floral volatiles and interactions with herbivores and pollinators.

Volatiles Mediate Plant Interactions With Natural Enemies and Pollinators

Volatile duality has been well studied for plants regulating interactions with insect herbivores and beneficial insects (see above). When, why, and how volatiles mediate plant interactions with pollinators and natural enemies is far less understood (Kessler 2015), despite the potential for additive or synergistic effects on crop yield through biological control and pollination services (Gagic et al. 2019). Here, we summarize what is known about how volatiles shape plant interactions with these two groups of agriculturally important insects. We define “positive” outcomes as instances of increased natural enemy and pollinator attraction, likely leading to enhanced biological control and pollination services. We define “negative” outcomes as the opposite and focus on instances of decreased natural enemy and pollinator attraction or diminished ecosystem services (Fig. 3). However, it is important to note that certain factors increase the complexity of this framework. First, natural enemies and pollinators can respond to constitutive or herbivore-induced volatiles, which may be produced by leaves or flowers (Kessler and Halitschke 2007, Heil 2008, Lucas-Barbosa et al. 2011, Lehrman et al. 2013, Colazza et al. 2023). Rather than measuring insect responses to all possible combinations of such plant volatiles, studies have tended to focus on one type, resulting in a limited understanding of how beneficial insect behaviors change over plant ontogeny or with herbivory. Second, to the best of our knowledge, there are few studies that directly link volatile emissions to both natural enemy and pollinator behaviors, abundances, or their

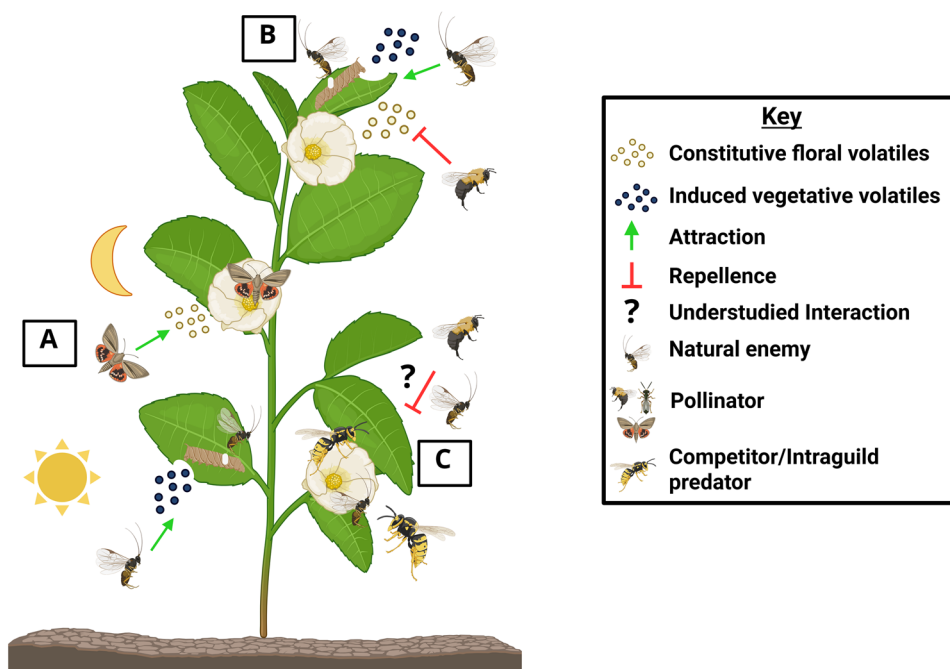


Fig. 3. Volatile-mediated plant interactions with insect natural enemies and pollinators depicting: (A) temporal separation of beneficial insects to maximize defense and reproduction, (B) plant trade-offs in defense and reproduction, and (C) avoidance of competitors or intraguild predators. The key identifies constitutive floral (light circles) or induced vegetative (dark circles) volatiles, insect attraction (pointed arrows) or repellence (blunted arrows), whether the interaction is understudied in the literature (question mark), and insect functional group (natural enemies or pollinators).

respective ecosystem services. Finally, simultaneous attraction of both insect groups may not always be positive. For instance, attraction of natural enemies and pollinators could result in predator consumption of pollinators or predator-avoidance behaviors in pollinators (Muñoz and Arroyo 2004, Dukas and Morse 2005).

Positive Outcomes—Attract Natural Enemies, Attract Pollinators

Strategic volatile emissions may allow plants to attract natural enemies and pollinators for indirect defense and reproduction, respectively. In many cases, natural enemies and pollinators share food resources (ie pollen and nectar) and habitats (Colazza et al. 2023), easily facilitating dual attraction. For example, green leaf volatiles in *Scrophularia umbrosa* (Loisel.) flowers attract social wasps for pollination and also activate indirect defense against caterpillars (Schiestl 2015). However, attraction of natural enemies and pollinators can lead to competition for shared floral resources (Jeavons et al. 2023). For example, the presence of nectar-foraging ants on pumpkin flowers reduced honey bee visitation and resulted in about 17% of honey bees being preyed upon (Sinu et al. 2017). To overcome possible challenges from dual attraction, plants can spatially or temporally separate natural enemy and pollinator recruitment through volatile emissions. For instance, *Acacia* plants—which form strong mutualisms with predatory *Acacia* ants—repel ants from young flowers by releasing *E*, *E*- α -farnesene, which allows pollinating bees to visit flowers and facilitate reproduction without the risk of predation (Willmer and Stone 1997, Willmer et al. 2009). Moreover, in *N. attenuata*, the terpene (*E*)- α -bergamotene is released from caterpillar-damaged leaves during the day to attract predators (Kessler and Baldwin 2001), as well as by flowers at night to attract nocturnal pollinators (Zhou et al. 2017). In the case of these wild plant species, the evolution of volatile blends is likely influenced by the strength of different mutualisms formed with insects, which may allow plants to balance defense and reproduction (Hojo 2022). For crop plants, volatile emissions are affected by a much wider range of factors, including breeding (Benrey 2023) and co-evolutionary history with a pest (Thompson et al. 2025a, 2025b), which may complicate volatile-mediated attraction of pollinators and natural enemies, particularly in instances of separate recruitment for each insect group. However, spatial or temporal separation of beneficial insect recruitment in agroecosystems may not always be necessary. For example, syrphid flies—*Eupeodes corollae* (Fabricius) and *Sphaerophoria rueppellii* (Wiedemann)—consume aphids (*Myzus persicae* Sulzer) as larvae and pollinate sweet pepper plants (*Capsicum annuum* L.) as adults, thereby increasing seed set and fruit production through both ecosystem services (Pekas et al. 2020).

To enhance ecosystem services, several studies have screened different crop plant species or varieties for volatile emissions that increase natural enemy or pollinator attraction. For example, HIPVs and subsequent parasitoid attraction were quantified for 146 maize genotypes to identify highly attractive volatiles and genotypes (Tamiru et al. 2020). A similar strategy has been employed to understand volatiles associated with pollinator visitation—for instance, several onion varieties emit dioxolanes and organosulfurs, which decreased bee recruitment and seed production (Soto et al. 2015). Other approaches to enhance pollination include identifying ubiquitously attractive

volatiles across several plant species (Dekebo et al. 2022). For example, apricot (*Prunus armeniaca* L.), cherry (*Prunus avium* L.), plum (*Prunus persica* (L.) Batsch), avocado (*Persea americana* Mill.), carrot (*Daucus carota* var. *sativus* (Hoffm.) Schübl. & G. Martens), radish (*Raphanus sativus* L.), and cabbage (*Brassica rapa* L. subsp. *chinensis*) emit linalool and phenylacetaldehyde, which strongly elicit behaviors associated with honey bee feeding (eg proboscis extension; Mas et al. 2020). In addition to cash crops, companion plants or cover crops can also produce highly attractive volatile emissions (Géneau et al. 2013). For example, You et al. (2024) tested 44 different flowering plant species that could serve as companion plants and found that *Fagopyrum esculentum* (Moench) produced floral volatiles—diethyl benzene and 1,4-diethyl benzene—which were highly attractive to *Cotesia* parasitoids. Despite these exciting results, no studies, to the best of our knowledge, have screened crop species or varieties for simultaneous attraction of natural enemies and pollinators. This may be challenging, as volatiles can be emitted by different plant structures (vegetative or reproductive) at the same or different times. However, the potential benefits are great, as natural enemies can protect plants from damage by herbivores, while pollinators can increase yield in many crops (Sutter and Albrecht 2016).

Negative Outcomes—Repel Natural Enemies, Repel Pollinators

Plant volatile emissions may repel both pollinators and natural enemies. In the case of pollinators, it may be beneficial for plants to emit repellent volatiles to prevent herbivory or nectar robbing (Farré-Armengol et al. 2013). Repellent floral volatiles may also play a role in enhancing the probability of successful reproduction, as plants can repel less-effective pollinators and attract highly specialized ones (Schmidt 1982, Junker and Blüthgen 2010). In the case of natural enemies, plants may emit repellent volatiles to minimize interference with reproduction. For example, predatory bugs and spiders are known to prey on pollinators (Dukas and Morse 2005, Hernandez et al. 2019), suggesting plants may benefit from repelling natural enemies under certain scenarios. Since few studies concurrently measure natural enemy and pollinator activity (Shackelford et al. 2013, Jeavons et al. 2023), there is limited evidence that plants emit volatiles to simultaneously repel both insect groups. Such volatile-mediated repellence could decrease both biological control and pollination services in agroecosystems, but it has not yet been investigated.

For plants, attraction of one beneficial insect group and repellency of the other may lead to trade-offs in defense and reproduction. To navigate such trade-offs, plants can dynamically leverage volatile emissions. For example, *B. rapa* plants infested with *Pieris* caterpillars can prioritize defense over reproduction by reducing floral volatile emissions, subsequently decreasing pollinator attraction and increasing parasitoid attraction (Schiestl et al. 2014). Pollinators can also learn to avoid areas where predators are present, leading to decreased visitation and yield (Romero et al. 2011). Alternatively, natural enemies could also avoid habitats with intraguild predators (Cotes et al. 2015) or conspecifics (Janssen et al. 2002), potentially decreasing abundances of natural enemies that prey upon or parasitize herbivores. Additionally, it is possible that floral volatiles may “mask” attractive HIPVs emitted by leaves,

which could impact the behaviors of pollinators and natural enemies. However, little is known about how vegetative and floral volatiles interact with one another (Holopainen and Blande 2013), as well as how such chemical interactions could impact beneficial insects. For example, *B. rapa* floral volatiles, including nonanal, decanal, phenylacetaldehyde, acetophenone, *p*-anisaldehyde, and α -farnesene, decrease attraction of *Cotesia* parasitoids to leaf HIPVs by up to 73% (Desurmont et al. 2015, 2020). Ultimately, understanding which ecological processes plants prioritize in volatile-mediated trade-offs offers opportunities to find ways to bolster specific ecosystem services for crop plants.

Plant Volatiles for Insect Management in Agroecosystems: Future Directions

Plants rely on volatile emissions to regulate interactions with insects, including herbivores, natural enemies, and pollinators. Here, we reviewed current literature on volatiles mediating plant interactions with herbivores and natural enemies, herbivores and pollinators, and natural enemies and pollinators. Despite substantial progress in assessing how plant volatiles influence agriculturally important insect groups, we uncovered several key knowledge gaps. First, future research in agroecosystems should take a community-level approach by investigating concurrent effects of plant volatiles on herbivores, natural enemies, and pollinators. Second, more detailed studies are needed on the spatial and temporal dynamics of plant volatile emissions, as well as how these volatile dynamics correspond to plant and insect phenology. Third, abiotic stressors and anthropogenic factors can alter plant volatile emissions, but how such changes in volatiles disrupt the behaviors of agriculturally important insects remain relatively unknown. Lastly, we call for collaborative efforts between entomologists and crop breeders to disentangle ecological and genetic mechanisms underlying volatile emissions that affect beneficial insects to ultimately enhance ecosystem services. Overall, we are poised to advance our knowledge on how plant volatiles regulate insect behaviors and discover new semiochemical tools for sustainable insect management in agroecosystems.

Crop plants emit volatiles to regulate ecological interactions with a myriad of insects, facilitating both positive and negative outcomes (Figs 1–3). However, to our knowledge, there are no studies that have simultaneously examined behaviors of herbivores, natural enemies, and pollinators in response to plant volatile emissions. Implementing a community-centric approach to study plant volatiles will shed light on their multifunctionality, providing deeper insights that are not gleaned from studying one or two insect groups alone (eg Wilby and Thomas 2007, Vandermeer et al. 2019). Plant volatiles and insects do not necessarily have a one-to-one relationship, meaning individual volatiles or blends of many chemicals can provide distinct and meaningful information to single or multiple insect groups (ie Pallini et al. 1997, Kappers et al. 2010, Kessler et al. 2010, Page et al. 2014, Schiestl 2015). By studying entire insect communities interacting with plants, we can better understand how plant volatiles function in mediating direct and indirect multitrophic interactions. This holistic view is necessary for developing sustainable pest management strategies that work with, rather than against, the natural ecological processes occurring in agroecosystems.

To determine how insect communities respond to plant volatile emissions, we propose future research focuses on linking laboratory assays to field studies. In a laboratory setting, methodologies are established for characterizing plant volatile emissions and examining detailed behaviors of specific insect species (Tholl et al. 2006, Park et al. 2019). However, a laboratory setting often neglects the complex interactions between plants and their associated insect communities. Pairing laboratory experiments with field studies can uncover unexpected insect behaviors and community dynamics in real-world agricultural settings. A powerful tool for researchers to understand the link between insect communities and plant volatiles is the use of bipartite network analyses (Burkle and Runyon 2019, Kantsa et al. 2019). Bipartite network analyses allow researchers to visualize interaction networks between plants and insects, as well as quantify the relative strength of these interactions. We propose applying bipartite analyses to research on agricultural crops through observational field studies of insect communities on different crop cultivars or varieties that produce unique volatile blends (ie chemotypes). Interactions between chemotypes and insect communities can then be converted into interactions between volatile compounds and insect communities, as shown in the hypothetical example in Fig. 4. Generating such bipartite networks will begin to reveal insect community-wide responses to plant volatile emissions and may elucidate compounds that are important for regulating behaviors among different insect groups.

An insect community-centric approach may also advance established insect management strategies, including push–pull systems and synthetic volatile lures. Push–pull systems manage insect pests by arranging cash crops nearby other plants that emit repellent and attractive volatiles, “pushing” and “pulling” pests away from cash crops, respectively (for reviews on push–pull systems, see Pickett et al. 2014, Eigenbrode et al. 2016, Zhou and Jander 2022). In many African countries, push–pull systems have been shown to increase crop yield while decreasing insect pest damage and agrochemical inputs (Martel et al. 2005, Ndayisaba et al. 2020). Push–pull systems are most effective in small agricultural fields (50 m × 50 m plots; Khan et al. 2019), but conventional farms in the United States are predominantly monocultures and substantially larger than recommended field sizes for successful push–pull application (USDA-NASS 2024). However, there are increasing calls for diversifying habitats in conventional farms to enhance economic and environmental resilience (Prokopy et al. 2020). When diversifying habitat in conventional farming systems (eg planting cover crops or meadows for pollinators), consideration of attractive and repellent plants—akin to classic push–pull systems—may increase the efficacy of habitat diversification.

Furthermore, synthetic lures augment volatile compounds present in agricultural fields, often to attract natural enemies for enhanced biological control and, in some cases, repel insect pests (Williams et al. 2008, Mallinger et al. 2011, Kelly et al. 2014, Athanasiadou and Meyhöfer 2025). To date, most lures emit single compounds or simplified volatile blends (Nielsen et al. 2019, Kansman et al. 2023). For example, lures containing methyl salicylate—a compound known to attract many natural enemies—and (*E*, *E*)- α -farnesene not only increased abundances of stink bug parasitoids but also increased egg parasitism rates by up to 80% in soybean (*Glycine max* (L.) Merr.) fields (Michereff et al. 2024). In some cases, simplified compounds may be sufficient to enhance agroecosystem services,

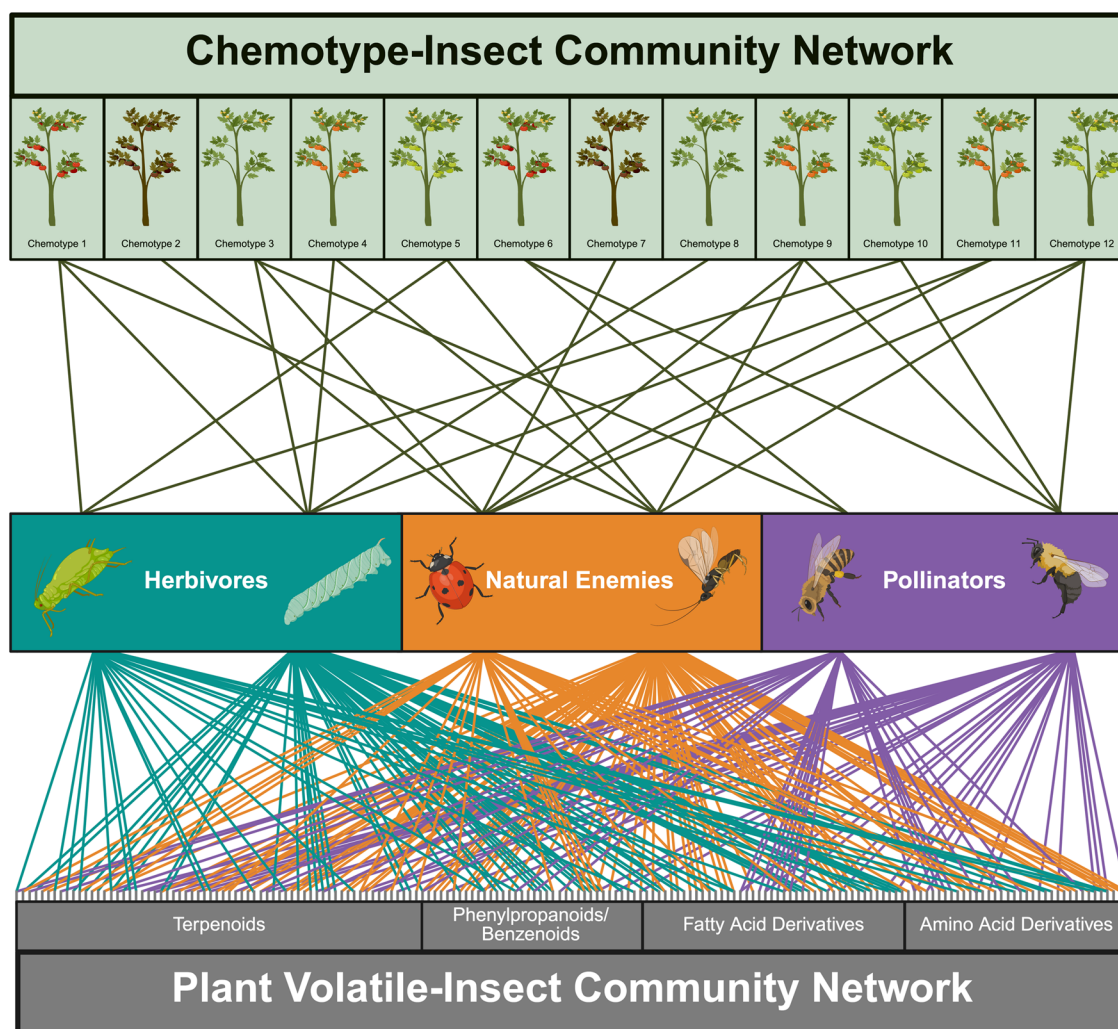


Fig. 4. Conceptual figure showing simplified insect community and how to link insect herbivore, natural enemy, and pollinator behaviors to plant volatiles emissions. Using different crop plant chemotypes, the chemical blends of chemotypes can be converted into linkages between plant volatiles and insect communities. Tick marks above volatile chemical classes represent individual compounds.

but in many cases, more complex and dynamic volatile blends would likely manipulate a wider range of insects. Applying knowledge gained from an insect community-centric approach could result in the formulation of synthetic volatile lures tailored to different crop systems, targeting the primary pest of concern while also attracting effective natural enemies and pollinators. For example, specialist natural enemies rely on more specific volatile blends to find insect hosts than generalists (Van Oudenhove et al. 2017). Elucidating these specific volatiles, as well as cross-attraction with pollinators or repellency of target and nontarget pests, will strengthen the use of synthetic volatile lures in agroecosystems.

In addition to adopting an insect community-centric approach, another critical factor to consider for insect control in agroecosystems is temporal variation in plant volatile emissions. Volatile emissions change before and after important phenological stages for plants, which can influence the behaviors of different insect groups. Pollinators home in on floral volatiles and use changes to floral volatiles to guide them to unpolinated flowers, which optimizes plant reproduction and provides pollinators with the ability to maximize floral resources in fresh, less-visited flowers (Rodríguez-Saona et al.

2011). Frugivorous pests and seed predators may too home in on floral volatiles, or the lack thereof, as the end of flowering signals the start of fruit and seed production, their preferred food sources (Rodríguez et al. 2013). In conjunction with temporal variation in plant volatiles, responding insects also grow and change over time. The effectiveness of volatiles in repelling or attracting insects can vary depending on insect life stages. For instance, volatiles that successfully repel pests as larvae may have no impact on adults, which could be problematic if both larvae and adults are crop pests (ie striped cucumber beetles; Andrews et al. 2007). By incorporating temporal variation in plant volatiles, chemical ecologists can design more sophisticated insect management tools that are deployed and activated to coincide with peak activity times of the target insects, maximizing volatile effectiveness and ultimately reducing nontarget impacts.

Future research on how plant volatiles regulate insect behaviors should also consider abiotic stressors and anthropogenic factors (for a review of how anthropogenic factors affect plant volatile emissions, see Blande et al. 2014, Bergman et al. 2025). Extreme temperatures are a major abiotic stressor for plants that can impact associated insects. For instance, when ambient

temperatures increase, pollinators (*B. terrestris*) and ovipositing butterflies (*Pieris rapae* L.) are repelled by heat-induced changes in *Brassica rapa* floral volatiles (Trainee et al. 2024). Similarly, increased temperatures enhance caryophyllene emission from potato (*Solanum tuberosum* L.) plants, repelling herbivores (*Phthorimaea operculella* Zeller) but enhancing parasitoid (*Trichogramma chilonis* Ishii) attraction to host plants (Munawar et al. 2022). Together, these examples suggest increasing temperatures alter volatile-mediated plant-insect interactions, although broader patterns across a greater diversity of crop plants and insects remain unknown. Anthropogenic factors can also modify plant volatile emissions and subsequently alter insect behaviors. For instance, high levels of carbon dioxide reduce HIPV emissions in maize (*Z. mays*), which may impact different insect groups, including natural enemies that typically navigate towards HIPVs (Block et al. 2017). Ozone is another important gas influenced by anthropogenic factors. Ozone alters plant volatile blends and disrupts volatile emission into the atmosphere, inhibiting chemical cues for foraging insects, especially in urban areas (reviewed by Masui et al. 2021). Overall, prior studies on abiotic stressors or anthropogenic factors impacting volatile-mediated plant-insect interactions have primarily focused on a single factor. However, in nature, many stressors and factors are often acting in concert, and future studies should consider investigating multiple, simultaneous abiotic stressors or anthropogenic factors to better understand their effects on plant volatile emissions, and in turn, insects in agroecosystems.

To breed crop varieties for enhanced ecosystem services, interdisciplinary studies are needed that link genetic mechanisms to ecological impacts of plant volatiles. Although genetic engineering of crop cultivars has been proposed as a promising avenue (Kos et al. 2013, Tamiru et al. 2020, Russavage et al. 2025), this approach comes with some pitfalls. For instance, genetic engineering for modifications to constitutive volatile emissions can increase attraction of herbivorous pests and reduce yield (Robert et al. 2013). Traditional crop breeding methods (eg crossing and selection) may produce varieties that strategically emit volatiles and increase beneficial insect attraction, but there is limited work in this area, possibly for a few different reasons. First, complex genetic factors may impact plant volatile biosynthesis, including pleiotropy (eg single terpene synthase gene produces several different volatiles) or genetic linkage (eg close association of different terpene synthase genes on a chromosome, Zu et al. 2016, Xu et al. 2020). As a result, predicting patterns of inheritance and volatile expression may be complicated (Stenberg et al. 2015). Second, volatile emissions exhibit phenotypic plasticity and are strongly affected by environmental factors (Ashra and Nair 2022), such as temperature (Trainee et al. 2024) or drought (Campbell et al. 2019). Such plasticity in volatile emissions makes crop phenotyping and breeding challenging. Third, there is no established framework for determining relationships between plant genes, volatiles, insect behaviors, and ecosystem services. Volatile collections and behavioral assays are commonly employed in laboratory settings to identify crop varieties that emit attractive volatiles (Tholl et al. 2006, Poelman et al. 2009, Russavage et al. 2024). However, this approach does not establish whether volatiles are still attractive under field conditions (Degen et al. 2012, Russavage et al. 2025), as well as what the effects are on other insect groups or what genes underlie volatile emissions. Moreover, the identification and development of molecular

markers associated with volatiles could expedite breeding and increase the precision of trait selection (Tamiru et al. 2015, 2020). An integrative approach across entomological disciplines (ie biological control, pollination ecology, chemical ecology) and plant genetics is needed to accomplish this goal.

Overall, we conclude that plant volatiles mediate complex interactions with multiple groups of insects, including herbivores, natural enemies, and pollinators. Plant volatiles have potential to be agriculturally useful in managing pests and promoting beneficial insect activity. Most research to date has focused on the role of volatiles in pest management, but studies indicate natural enemies and pollinators are also responding to volatiles in nuanced ways and should be further studied, especially in crop species, and from the perspective of an insect community. Bipartite networks may offer a useful framework for studying and understanding plant volatile-mediated interactions at the insect community-scale. Future research should also consider how temporal variation, abiotic stressors, anthropogenic factors, and genetic engineering are affecting plant-insect interactions.

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Conflicts of interest

None declared.

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